Data Structures and Algorithms (CS1.201)

Mid-Sem Exam (27 Feb, 2026)
Total Marks: 30 [10 + 12 + 8]
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ATTENTION! *Please check that this question paper has* **3 puzzles**, *which add to* **30 marks** *total. All the best!* ☺

Puzzle 1 [2 + 2 + 2 + 2 + 2 = 10 marks]. Consider a min-heap with the following additional property (called property P):

Property P: For every node, all elements in its left subtree are strictly smaller than all elements in its right subtree.

Answer the following questions, providing a proper and unambiguous logical reasoning. (Assume that all of the elements of the heap are distinct.)

1. Given a min-heap having property P, which tree traversal (preorder, inorder, postorder) will output all of its elements in a strictly increasing order?
2. We know that, in general, there could be many different min-heaps for a given set of elements. How many different min-heaps satisfying property P can a fixed set of elements have?
3. Suppose a new element that is strictly larger than all of the existing elements is inserted into a min-heap having property P. Explain where it must eventually reside.
4. If we apply the standard bottom-up heap construction algorithm to an arbitrary array, will the resulting min-heap necessarily satisfy property P?
5. In a min-heap having property P, if the root is deleted using the standard heap deletion method (replace with last element and sift down), then will the resulting min-heap still have property P?

Puzzle 2 [2 + 1 + 4 + 5 = 12 marks]. Let $A[1, \ldots, n]$ be an array of n distinct numbers. If $i < j$ and $A[i] > A[j]$, then the pair (i, j) is called an inversion in A.

1. List all the inversions in an array A of size 5, given $A[1] = 2,\ A[2] = 3,\ A[3] = 8,\ A[4] = 6,\ A[5] = 4$.
2. Which array of size n with elements from the set $\{1, 2, 3, \ldots, n\}$ has the most inversions?
3. What is the relationship between the running time of insertion sort and the number of inversions in the input array? Justify your answer.
4. Give an efficient algorithm to count the number of inversions in an array.

Puzzle 3 [4 + 2 + 2 = 8 marks]. A group of n people are standing in a circle, and are numbered from 1 to n in clockwise order. They have an elimination procedure, which goes as follows:

Elimination Procedure: Starting from person **1**, *every person (who is not yet eliminated) will eliminate the first person standing next to them in the clockwise direction (a person once eliminated is completely ignored for the rest of the elimination procedure). This continues until only one person remains. In other words, the elimination procedure ends when $(n-1)$ persons have been eliminated.*

For example, if $n = 11$ (see Figure 1 below), then the following sequence of operations will take place in the elimination procedure:

1. Person 1 eliminates person 2
2. Person 3 eliminates person 4
3. Person 5 eliminates person 6
4. Person 7 eliminates person 8
5. Person 9 eliminates person 10
6. Person 11 eliminates person 1
7. Person 3 eliminates person 5
8. Person 7 eliminates person 9
9. Person 11 eliminates person 3
10. Person 7 eliminates person 11

In this manner, after 10 eliminations have been done, only one person remains. And therefore, the sole survivor in this example is person 7. Your task is to design an algorithm for n people to determine:

- The order in which people are eliminated.
- The last remaining person.

Your algorithm will take as input a positive integer n, and it will output a sequence showing the order of their elimination, as well as who the sole survivor is. Your solution will be graded on the basis of (the correctness of) its code/pseudo-code, (the analysis of) its running time, and (the clarity of) its explanation.

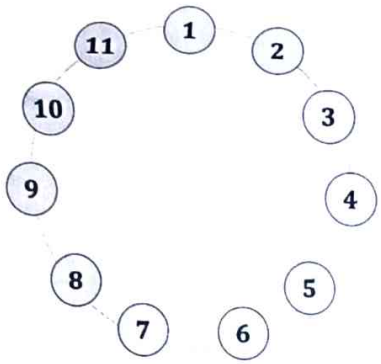

Figure 1: The initial arrangement of the 11 people